

BUSINESS INSIGHT REPORT

1. Introduction

The analysis investigated the factors associated with heart disease using different variables. It included advanced exploratory data analysis, statistical analysis, and feature engineering. The main objective was to identify variables that could provide useful predictive information for a future machine-learning model.

2. Key EDA Findings

The exploratory analysis identified differences in the distribution of several variables between patients with and without heart disease. Particular attention was given to age, maximum heart rate, Oldpeak, chest pain type, exercise-induced angina and ST slope. The class distribution was also examined to determine whether the target variable presented a significant imbalance.

3. Statistical Findings

The statistical analyses performed were:

- *Pearson correlation*
- *Chi-square test*
- *Welch independent t-test*
- *One-way ANOVA*

The statistical results were interpreted using a significance level of 0.05. Statistically significant relationships were identified where the p-value was below 0.05.

4. Feature Engineering

The engineered variables created were:

- *AgeGroup*
- *BP_Category*
- *Cholesterol_Risk*
- *MaxHR_Category*
- *Exercise_Stress*

These features were created to represent non-linear relationships and clinically interpretable categories. The engineered features will be evaluated further during machine-learning model development.

5. Feature Selection

Features were evaluated using correlation analysis, statistical significance, categorical association, and mutual information. Features demonstrating meaningful predictive information were retained for modelling.

6. Business Recommendations

The analysis suggests that data-driven risk assessment can help identify patient characteristics associated with increased likelihood of heart disease. The most informative variables identified through statistical analysis should receive particular attention during predictive model development. ***However, the analysis identifies statistical associations rather than causal relationships and should not be interpreted as a clinical diagnostic system.***

7. Tools to be Used

Python - Primary programming language for data analysis and experimentation.

Google Colab- To organize and document the Week 3 analysis.

Pandas - For data loading, cleaning, transformation, aggregation, descriptive statistics, and data manipulation.

NumPy - For numerical operations and mathematical calculations.

Matplotlib - To create customized statistical and exploratory visualizations.

Seaborn- For distribution plots, boxplots, categorical analysis, correlation heatmaps, and other statistical visualizations.

SciPy- For statistical hypothesis testing, including t-tests and chi-square tests.

Scikit-learn- For machine-learning-based feature importance and supporting feature analysis.

GitHub- To store, document, and share the project repository. Analytical Techniques